MILITARY SPECIFICATION

MANUALS, TECHNICAL; TAPES; AND CARDS
(Programmed Test Equipment)

1. SCOPE

1.1 This specification covers the preparation of General Procedures Manuals, Test Procedures Manuals, program tapes, and checkout cards, all of which are used in conjunction with programmed test equipment.

1.2 The General Procedures Manual will serve as a guide for using a series of Test Procedures Manuals which contain specific test procedures for tape and card programmed test equipment.

1.3 Normally, a Test Procedures Manual supports each program tape or deck of checkout cards on how the cards or tape are to be used and how to interpret the indications.

1.4 Program tapes and cards are used in operating special device equipment resulting in immediate analysis and performance of an operational system or operation of airborne computers containing information causing or enabling guidance systems to function. The program tapes and cards provide test data required for programmed test equipment. These programs will be developed for organizational, intermediate, or depot level tests.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for

PUBLICATION

Technical Order

00-5-1	Air Force Technical Order System

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preparation of manuals. The general manner of preparation shall be in accordance with MIL-M-38784 and MIL-P-38790. Manuals shall not repeat information normally required in operation, organizational maintenance, intermediate maintenance, or overhaul manuals, unless such repetition is deemed necessary for clarity or continuity, or both. Where repetition is required, the information shall be extracted in condensed form.

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3.1.1 Contents of general procedures manual. This manual shall include a general description of the programmed test equipment program (a program may relate to a weapon system or a family group, such as ground radars or airborne computers). The manual shall cover procedures which apply to the entire program; i.e., handling of tapes and cards, marking of tapes and cards, and storage of tapes and cards. If special tape or card handling procedures are required for individual testers, these procedures shall be described adequately to enable satisfactory tester operation. The general procedures manual shall include a cross index of tapes/cards to system or equipment, and system or equipment to tapes cards. The index shall provide the information required for maintenance personnel to assemble the manuals, tapes, or checkout cards and testers necessary for performing specific testing operations.

3.1.1.1 Front matter. The title page, list of effective pages, table of contents, and list of illustrations shall be prepared in accordance with MIL-M-38784.

3.1.1.2 Introduction. In addition to the requirements of MIL-M-38784, the introduction shall give the purpose of the manual and any relevant information that will be of assistance in its use. The introduction shall describe the purpose of the Test Procedures Manuals and how such manuals are to be used.

3.1.1.3 Text.

3.1.1.3.1 General information. A general information section shall be provided containing a brief summary of the test program. Specific items of equipment shall be mentioned only for purposes of illustration. The information contained in the discussion shall be kept at the general familiarization level. Theory of programmed test equipment, if any, shall be covered only briefly and shall be simple and general enough to be understood by maintenance personnel.

3.1.1.3.2 General maintenance instructions. This section shall contain general maintenance instructions which apply to program tapes and checkout cards. Where maintenance instruction apply only to tapes or cards, for specific systems or equipment, the instructions shall be mentioned in this section, but shall be discussed in detail in the applicable system section. Maintenance information shall include the procedures contained in the following paragraphs:

a Tape handling procedures. Explanation of the general procedures for handling program tapes, including the correct method of installing a tape on a reel, or into a magazine to insure proper test equipment and tape operation, and repair limitations, if any.

b. Tape marking. Explanation of the procedures for tape labeling, including label fabrication and the procedures for marking tapes other than by labels, as applicable. The approved materials to be used shall be specified.

c. Tape splicing procedures. Explanation of the procedures for splicing tapes, including magazine type tapes. The approved materials to be used shall be specified.

d Checkout card handling procedures. Explanation of the general procedures for handling checkout cards. Card defects caused by misuse or improper handling shall be defined.

e Card marking. Explanation of the procedures for labeling or marking checkout card decks and individual cards.

3.1.1.3.3 System sections. There shall be one section for each system which is checked by tape or card programmed test equipment. These sections shall contain a list of testers that utilize programmed tapes and cards, a cross index of tapes/cards to system or equipment, and a cross index of system or equipment to tapes/cards. Where no maintenance instructions are provided in a system section, the instructions in the general maintenance instructions section shall cover the requirement.

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3.1.1.3.3.1 Cross index listing of effective checkout tapes and cards. A cross index type listing of all applicable tapes, checkout cards, and programmed test procedures manuals versus the system, line replaceable unit or assembly to which they apply, together with a listing of adapters or special cabling required, shall be included. Checkout cards used in conjunction with program tapes shall be listed with the applicable tape. The cross index listing shall enable maintenance personnel to identify the instruction manual, program tape, and checkout card deck required to test a specific item or equipment. Samples of cross index listings follow:

Tape/Card Tech Order No.	Manual Tech Order No.	Tester Part No.	Adapter Part No.	Test Title	System or Equipment Checked
TO 21M-CGM16D-8-26-2CT-1	TO 21M-CGM16D-8-26-1	12345	67890	Test and calibration	Autopilot ground control unit, launch area checkout 29-802

System or Equipment Checked	Test Title	Tester Part No.	Adapter Part No.	Manual Tech Order No.	Tape/Card Tech Order No.
Autopilot ground control unit, launch area checkout 29-802	Test and calibration	12345	67890	TO 21M-CGM16D-8-26-1	TO 21M-CGM16D-8-26-2CT-1

3.1.1.4 Illustrations. Illustrations shall be provided to convey essential information or clarify the instructions in the text. Illustrations may appear on separate pages, but shall be placed as close to the associated text as possible. Illustrations shall conform to MIL-M-38784. Electrical and electronic wiring diagrams shall not be included.

3.1.2 Contents of test procedures manuals. Each manual shall contain all necessary information pertinent to connections, interconnections, cabling, and adapters; and procedures to position tester and equipment controls in the correct configuration to perform programmed test procedures contained on tapes and cards.

3.1.2.1 Front matter. The title page, list of effective pages, table of contents, and list of illustrations shall be prepared in accordance with MIL-M-38784.

3.1.2.2 Introduction. In addition to the requirements of MIL-M-38784, the introduction shall give the purpose of the manual and any relevant information that will be of assistance in its use. A statement shall be included that test equipment hook-up, tape handling and installation instructions and checkout card information (as appropriate) are contained in the applicable operation, organizational maintenance, intermediate maintenance, or overhaul manual, and the applicable technical order number(s) shall be cited. Positive identification of the applicable tape and checkout card as to equipment configuration shall be shown in chart or tabular form.

3.1.2.3 Text. The instructions shall be arranged in appropriate sections, and shall include complete step-by-step procedures which will enable the operator to evaluate the performance of the equipment being tested. Instructions shall also be given for manually programmed tests, if applicable. Detailed instructions shall be given for locating and identifying the cause of malfunction. The information shall relate symptoms of malfunctions of a system, equipment, or component to an accessory, assembly, subassembly, plug-in unit, detail part, etc, compatible with the tape or checkout card. The step-by-step procedures (operational checkout analysis, or alignment) shall be keyed to operation of the test equipment by using the test indication number in lieu of a paragraph number.

3.1.2.4 Illustrations. Illustrations shall be provided to convey essential information or clarify the instructions in the text. Illustrations may appear on separate pages, but shall be placed as close to the associated text as possible. Illustrations shall conform to MIL-M-38784. Electrical and electronic wiring diagrams shall not be included.

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3.2 Preparation of tapes. A program tape shall provide all the test data required for a specific item of programmed test equipment, as determined by the procuring activity. Program tapes shall be of Mylar 1.000 inch wide and be 0.0020 inch minimum thickness to 0.0053 inch maximum thickness (see figure 1). The following tolerances shall be observed:

a. The feed holes shall be round with a diameter of 0.046 inches,+.002, -.001 inches. The feed holes shall be spaced 0.100 inches, ± .001 inches. In addition, the accumulated longitudinal error between centers of feed holes shall not exceed ± .009 inches within the span of 6.000 inches.

b. The code holes shall be round with a diameter of 0.072 inches, +.001, -.002 inches. The code holes shall be spaced 0.100 inches, ±.002 inches in the transverse direction and ± .003 inches in the longitudinal direction.

c. All holes punched in the tape shall center on the true intersection of the longitudinal and perpendicular transverse center lines. All measurements shall be conducted on a flat level surface with no tension applied to the tape.

3.2.1 Identification of tapes. Program tapes shall be identified by permanent printed or inked markings. White pressure sensitive labels (FSN 7530-082-2661 or equivalent) may be used when applied before punching so the sprocket holes are punched through the label. The markings shall be fully visible when the tape is wound on a reel. Identification shall consist of the security classification (if any), technical order number, tape title, and title security classification (if any), applicable system, and type test (see figure 1). Technical order numbers and security classification will be assigned by the procuring activity.

3.2.1.1 Program tapes shall have a sufficient leader to feed through a reader head and provide for tape identification on both ends of the tape. The beginning and the end of the tape shall be clearly indicated. The words START and END shall be placed on the tape to prevent searching beyond the programmed area of the tape (see figure 1).

3.3 Preparation of checkout cards. A deck of checkout cards shall provide all the test data required for a specific item of programmed test equipment, as determined by the procuring activity. Cards shall be prepared in the style of figure 2 and in the form (master copy or printed copy) specified in the contract. Master copy shall be on card stock. Cards shall be printed on one side only. (An introduction card and a table of contents card are not required.)

3.3.1 Lettering of cards. All printed matter appearing on the cards may be typewritten or mechanically lettered; the size of type used shall not be less than 6-point.

3.3.2 Security classification. The security classification of the assembled card deck will be specified by the procuring activity. Cards which are classified higher shall not be included in the deck.

3.3.3 Detail requirements.

3.3.3.1 Title card. The title card shall be prepared in accordance with figure 2. The title card overall dimensions shall be the same as the associated checkout cards and may exclude all index and key cuts.

3.3.3.2 List of effective cards. A list of effective cards shall be prepared each time a basic or revised card deck, or a changed card to a card deck is issued. The format and content of the list of effective cards shall be in accordance with figure 2. The list of effective cards overall dimensions shall be the same as the associated checkout cards and may exclude all index and key cuts.

3.3.3.3 Card deck publication number. The technical order number of a deck of checkout cards will be assigned by the procuring activity.

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3.3.3.4 Card numbering. The list of effective cards shall be numbered using capital letters in sequence. Commencing with the first punched card following the list of effective cards, all cards of a basic or revised deck shall be numbered consecutively using Arabic numerals. Cards added by a change, between cards in the card deck, shall be numbered by suffixing a capital letter for each added card, starting with A (example: 22A, 22B); cards added at the end of the deck shall continue with the basic consecutive numbering system. Deleted cards shall be noted only in the list of effective cards.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions of T0 00-5-1, MIL-M-38784, and MIL-P-38790 are applicable.

5. PREPARATION FOR DELIVERY

5.1 Manuals. Packaging, packing, and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

5.2 Tapes. All paper, Mylar, metalized Mylar, and magnetic tapes shall be packaged in cardboard, metal, or plastic containers, whichever is appropriate, for protection against damaging effects from environmental conditions. All tapes shall be mounted on reels and placed in metal or plastic containers for shipment to depots and field activities.

5.3 Cards. All card decks shall be packaged and shipped in cardboard or plastic boxes.

6. NOTES

6.1 Intended use. Refer to 1.1 through 1.4.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.

b. Whether master copy or printed copies of cards shall be provided(3.3)

6.3 Supersession data. This specification supersedes MIL-M-9889A(USAF) dated 10 March 1961, MIL-P-9899(USAF) dated 18 January 1961, MIL-T-9901(USAF) dated 17 February 1961, and MIL-C-9905(USAF) dated 27 March 1961.

Custodian:
Air Force - 16

Preparing Activity:
Air Force - 16

Review Activities:
Air Force - 01, 10, 70, 71, 80, 82, 84

Project TMSS-F331

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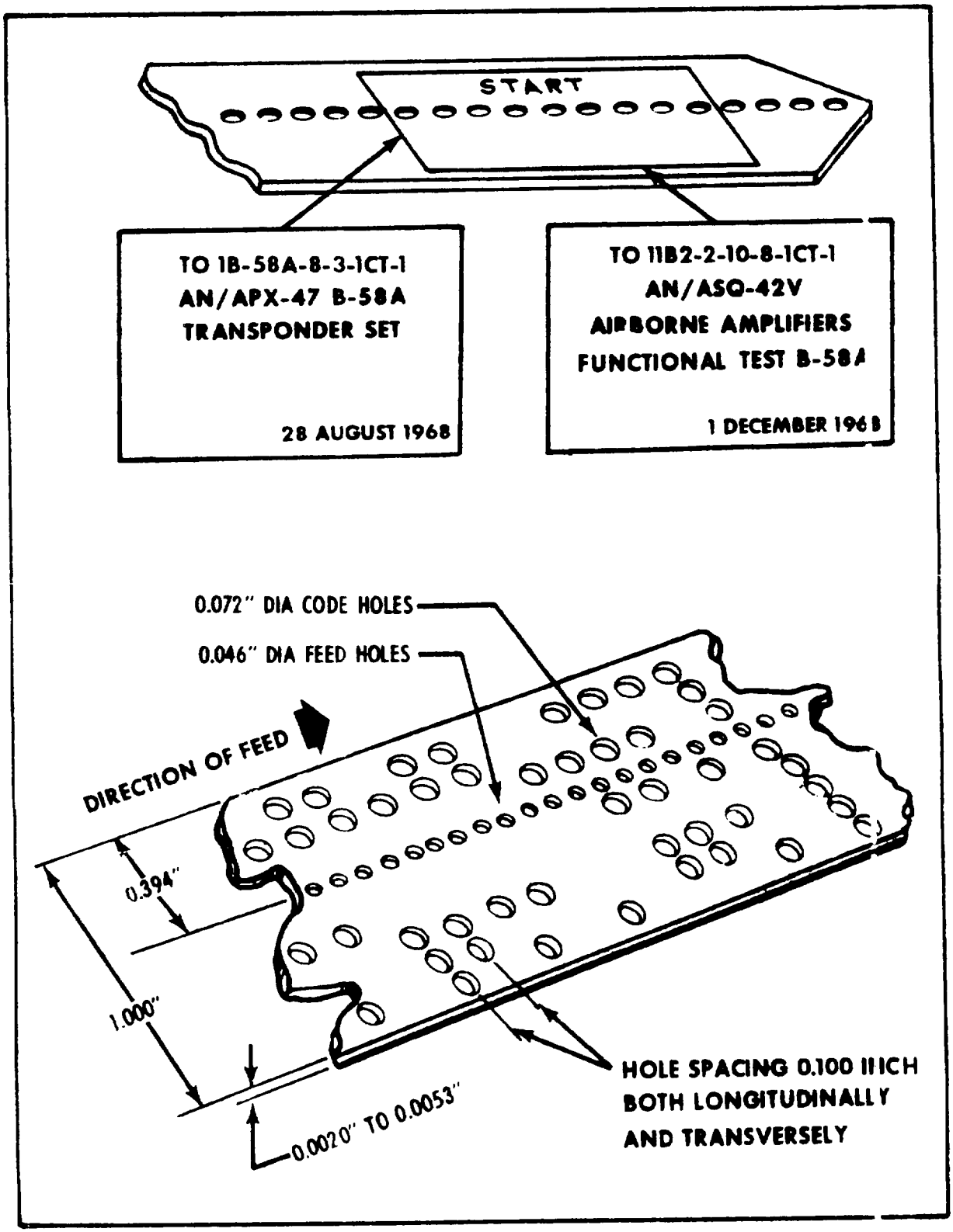


Figure 1. Program Tape. (Sheet 1 of 2)

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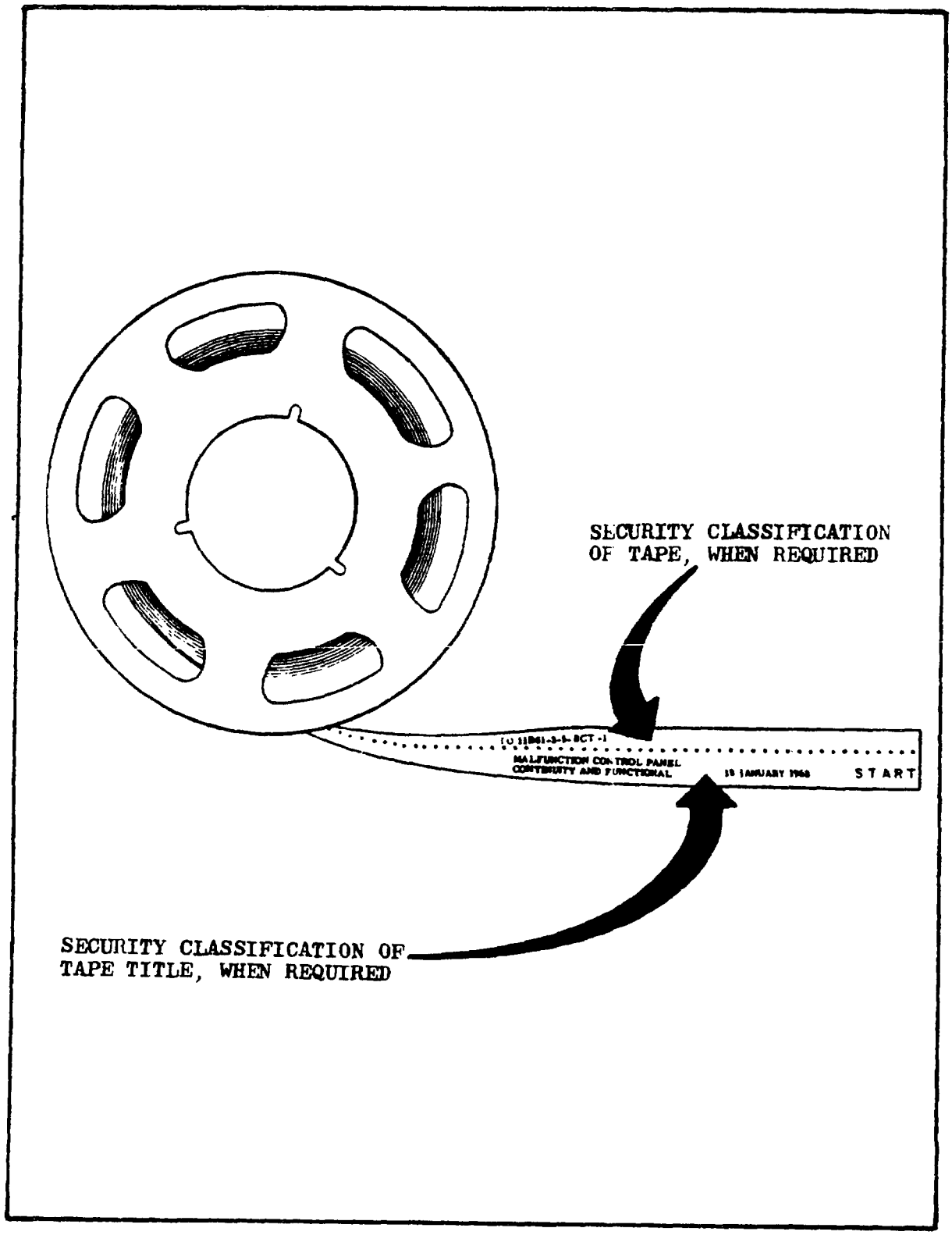


Figure 1. Program Tape. (Sheet 2 of 2)

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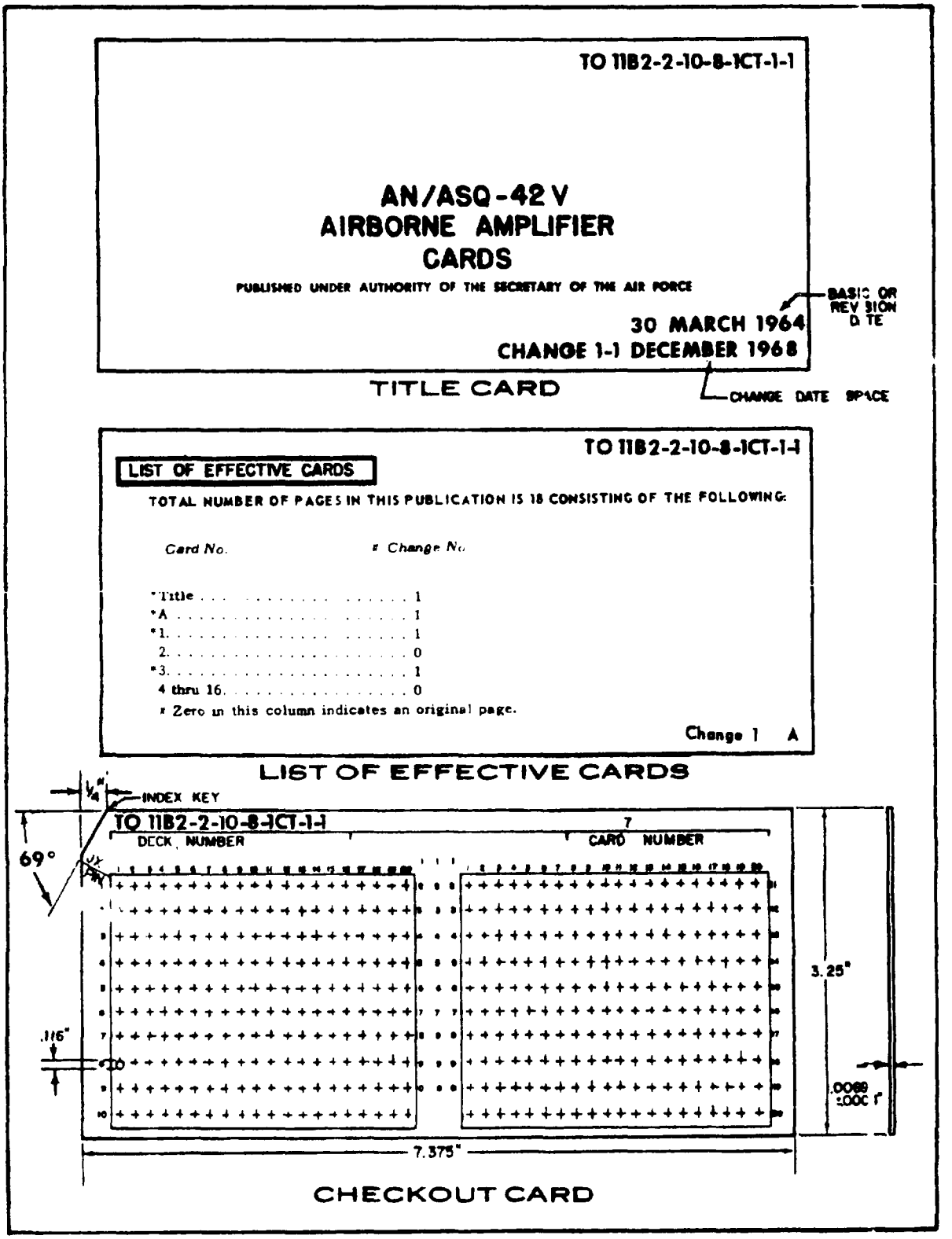


Figure 2. Checkout Card Deck

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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